Push 2 / Nuendo Bridge

Plugin Mapper Guide — Version 1.0

1. Overview

The Plugin Mapper lets you create custom parameter mappings for your VST3 plugins. When you focus a mapped plugin in Inserts mode, the Push 2 encoders automatically control the parameters you have assigned, organized in pages of 8.

- **Scanner**: discovers installed VST3 plugins and extracts their parameter lists using Spotify's pedalboard library (GPL-3.0).
- **Web interface**: a drag-and-drop UI served by FastAPI at http://localhost:8100 for creating and editing mappings.

The server is integrated into the bridge and starts automatically if dependencies are installed. It is entirely optional.

2. Installation

Install the required packages:

```
pip install fastapi uvicorn pedalboard
```

Without pedalboard (web interface only, no scanning):

```
pip install fastapi uvicorn
```

Startup messages:

- **running at http://localhost:8100, scanner ready** — everything installed
- **running, scanner unavailable** — pedalboard missing
- **not available** — fastapi/uvicorn missing (bridge works normally)

3. File Locations

File	Location	Description
Plugin cache	~/.push2bridge/plugin_cache.json	Scanned plugin parameters (auto-generated)
Mappings	~/.push2bridge/mappings/*.json	One JSON file per mapped plugin
Settings	~/.push2bridge/mapper_settings.json	Scanner settings (directories, auto-scan)
Mapper source	src/mapper/	Scanner, server, and web interface files

*All user data is stored in **~/.push2bridge/**. Mappings are plain JSON files that can be shared, backed up, or edited manually.*

4. Scanning Plugins

The scanner discovers VST3 plugins in standard directories:

- macOS: /Library/Audio/Plug-Ins/VST3/ and ~/Library/Audio/Plug-Ins/VST3/
- Windows: C:/Program Files/Common Files/VST3/

To scan, open the web interface and click Scan. Each plugin is loaded in an isolated subprocess with a 30-second timeout.

```
Command line: cd src/mapper && python scanner.py
```

Options: --force (rescan all), --retry (retry errors), --stats (cache statistics).

5. Creating Mappings

Open http://localhost:8100 or click Plugin Mapper in the macOS menu bar.

Left Panel — Plugin List

- Lists all scanned plugins with search and filter (All / Effects / Instruments / Mapped / Unmapped).

Center Panel — Parameters

- Shows all parameters of the selected plugin as draggable cards.

Right Panel — Mapping Pages

- Each page has 8 slots for the 8 Push 2 encoders.
- Drag parameters from the center panel into encoder slots.
- Click Add Page for additional pages (navigate with Left/Right on Push).
- Edit short labels for each slot (displayed on Push 2 screen).
- Click Save Mapping — saved to ~/.push2bridge/mappings/{plugin_name}.json.
- Click Delete Mapping to remove an existing mapping.

6. How Mappings Work

When you enter the parameter view for an insert plugin:

- 1. The bridge checks ~/.push2bridge/mappings/ for a matching JSON file.
- 2. If found, it asks Nuendo (via DirectAccess) to enumerate all plugin parameters.
- 3. It matches pedalboard parameter names to DA names using fuzzy word matching (e.g. 'band_1_frequency' matches 'Band 1 Frequency').
- 4. Configures the 8 encoders to control the matched parameters directly.
- 5. Displays parameter names and live values on the Push 2 screen.

If no mapping exists, the bridge uses Nuendo's default parameter bank.

> *Mappings are loaded at startup. Restart the bridge after creating or modifying mappings.*

7. Troubleshooting

- **Web interface not loading**: check that fastapi and uvicorn are installed.
- **Plugin not in list**: click Scan. Some plugins are incompatible with pedalboard.
- **Parameters don't match**: edit the mapping JSON manually if fuzzy matching fails.
- **Encoders don't respond**: ensure DA insert cache is built. Re-enter the insert to re-trigger.

8. Dependencies

Package	Version	License	Purpose
pedalboard	>=0.9.0	GPL-3.0	VST3 plugin loading and parameter extraction

fastapi	>=0.104.0	MIT	REST API server
uvicorn	>=0.24.0	BSD-3-Clause	ASGI server for FastAPI

All dependencies are GPL-3.0 compatible.